

SHIPU, China

WMO: 585690

Lat: 29.2047N

Lon: 121.9557E

Elev: 127

StdP: 99.81

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.6	0.6	-11.5	1.4	1.6	-9.1	1.8	2.5	11.2	6.2	10.1	6.3	4.9	340	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	32.6	27.2	31.5	26.8	30.5	26.5	28.1	31.0	27.5	30.3	27.0	29.6	4.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.3	23.5	29.7	26.7	22.6	29.3	26.2	21.9	28.8	91.1	31.1	88.4	30.4	85.9	29.8	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	9.9	8.8	DB	-2.5	35.0	1.8	1.3	-3.8	36.0	-4.8	36.8	-5.8	37.5	-7.1	38.5
			WB	-4.8	29.0	1.9	0.7	-6.2	29.6	-7.3	30.0	-8.3	30.4	-9.7	30.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	6.6	7.5	10.6	15.3	20.0	23.5	27.7	28.0	24.7	20.4	15.1	9.3
	DBStd	7.95	3.46	3.47	3.25	3.09	2.35	2.54	1.94	1.64	2.37	2.56	3.36	3.57
	HDD10.0	293	117	84	31	2	0	0	0	0	0	0	4	56
	HDD18.3	1413	365	303	239	101	10	0	0	0	11	104	281	
	CDD10.0	3009	10	14	51	159	311	404	549	557	441	322	157	33
	CDD18.3	1089	0	0	1	8	62	154	291	299	191	74	8	0
	CDH23.3	8367	0	0	1	11	118	773	2965	3117	1212	163	6	0
	CDH26.7	2291	0	0	0	1	8	143	919	978	232	10	0	0
Wind	WSAvg	4.6	4.7	4.7	4.6	4.4	4.1	4.3	5.1	4.9	4.8	4.9	4.5	4.7
Precipitation	PrecAvg	1467	73	77	125	117	137	222	108	190	184	84	88	62
	PrecMax	2020	202	152	318	269	279	400	286	392	365	264	220	175
	PrecMin	810	19	14	42	38	52	23	9	48	47	1	24	1
	PrecStd	258	41	37	67	52	66	86	68	91	89	68	53	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	19.6	21.3	24.8	27.7	31.5	34.5	34.2	31.5	27.9	24.2	19.2
		MCWB	15.1	15.8	15.9	19.1	22.4	27.0	27.6	27.5	26.8	24.0	20.3	16.3
	2%	DB	15.0	16.2	18.7	22.7	25.8	29.9	32.7	32.7	30.2	26.3	22.1	16.9
		MCWB	12.9	13.7	14.5	19.1	21.5	26.2	27.3	27.2	26.0	22.6	19.4	14.7
	5%	DB	13.2	14.0	16.8	21.3	24.6	28.3	31.5	31.7	29.1	25.0	20.9	15.5
		MCWB	11.1	11.8	14.1	18.4	21.1	25.3	26.9	26.8	25.3	21.5	18.5	13.2
	10%	DB	11.5	12.3	15.2	19.8	23.5	26.9	30.5	30.6	28.0	24.0	19.6	14.2
		MCWB	9.3	10.2	13.0	17.4	20.6	24.5	26.6	26.6	24.6	20.6	17.2	12.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.1	16.4	17.5	21.5	24.1	27.5	28.8	29.1	27.9	25.7	21.6	17.6
		MCDB	16.8	18.3	19.5	23.1	26.3	30.7	32.1	31.5	30.5	26.8	22.6	18.3
	2%	WB	13.6	14.2	15.8	20.3	23.0	26.5	28.1	28.1	26.7	23.9	20.5	15.4
		MCDB	14.6	15.9	17.5	21.6	24.5	29.1	31.3	30.8	29.0	25.2	21.6	16.4
	5%	WB	11.5	12.3	14.6	19.1	22.2	25.6	27.5	27.5	26.0	22.6	19.1	13.7
		MCDB	12.8	13.7	16.4	20.5	23.6	27.7	30.4	30.3	28.1	24.3	20.0	15.0
	10%	WB	9.7	10.6	13.3	17.9	21.4	25.0	27.0	27.0	25.4	21.5	17.8	12.3
		MCDB	11.0	12.0	15.0	19.4	22.8	26.6	29.6	29.7	27.4	23.3	19.2	13.9
Mean Daily Temperature Range	5% DB	MDBR	5.1	5.3	5.5	5.7	5.3	4.5	5.1	5.0	4.9	5.3	5.3	5.4
		MCDBR	7.3	7.9	7.9	7.6	6.8	6.1	6.8	6.7	6.0	6.3	6.5	6.9
		MCWBR	5.2	5.5	5.1	4.6	3.8	3.2	2.9	2.7	3.0	3.4	4.0	4.7
	5% WB	MCDBR	6.4	7.3	7.1	6.4	5.7	5.8	6.1	5.9	5.2	5.1	5.1	5.6
		MCWBR	5.3	5.7	5.2	4.6	3.9	3.2	3.0	2.9	3.0	3.5	4.0	4.7
Clear-Sky Solar Irradiance	taub	0.550	0.592	0.634	0.645	0.611	0.617	0.577	0.582	0.558	0.538	0.536	0.531	
	taud	1.772	1.711	1.657	1.690	1.815	1.852	1.976	1.949	1.975	1.920	1.868	1.823	
	Ebn at Noon	655	672	680	694	720	712	740	732	733	712	666	647	
	Edh at Noon	189	218	245	244	216	207	183	186	176	174	170	172	
All-Sky Solar Radiation	RadAvg	2.18	2.53	3.47	4.09	4.40	4.17	5.91	5.24	4.13	3.53	2.44	2.20	
	RadStd	0.36	0.59	0.33	0.46	0.56	0.54	0.60	0.51	0.47	0.49	0.34	0.37	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page